

The Combinatorics of Sibling Conflict: Why Family Size Creates an Exponential, Not Linear, Increase in Potential Points of Conflict

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August 2026

I have four children, each two years apart. I grew up with one sibling, so the dynamics of a larger family were new to me, and what kept striking me was that the number of distinct ways my kids could get into a fight seemed never-ending. So I set out to count it. The answer, for four children, is 25. In general, the number of ways n children can split into two opposing sides, with the remaining children uninvolved, is

$$\frac{3^n - 2 \cdot 2^n + 1}{2},$$

which is the Stirling number of the second kind $S(n+1, 3)$, and it grows exponentially with family size. Having the number changed how I read the noise in my own house, and it bears on something parents and researchers alike observe: larger families report more relational strain than smaller ones, and the strain seems to grow faster than the family does. The explanation on offer here is combinatorial, not psychological. Sociologists started counting family structure a long time ago; Bossard [2] counted relationships and Kephart [6] counted subgroups. What follows extends their arithmetic to opposing-sides conflict configurations and connects it to the modern literature on coalition structure generation in cooperative game theory. To be clear at the outset: I make no claim that this formula predicts how often real siblings fight, or how badly. It measures the size of the possibility space in which conflict can occur. My suggestion is that the size of that space, rather than any single behavioral trait, may be an underappreciated variable in family cohesion research.

1. The problem as commonly understood

Sibling relationship research has long documented that coalition formation is a real and recurring feature of family systems. Siblings form alliances to gain leverage against parents or against other siblings, and these coalitions shift over time. Structural family therapy treats the coalition, a joint action of two family members against a third, as a central unit of analysis

[9]. Caplow [3] showed that triads across social life, families included, tend naturally to divide into two-against-one formations. Contemporary reviews confirm that sibling dynamics operate within larger family system dynamics and help constitute them [8, 1]. All of this literature describes coalitions qualitatively, as social and emotional phenomena shaped by birth order, gender, and parenting style.

A quantitative tradition also exists, though it is older and largely dormant. Bossard [2] proposed a “law of family interaction”: with each person added to a family, the number of dyadic relationships grows as the triangular numbers, $n(n-1)/2$. Kephart [6] extended the count to possible subgroupings and showed that the combinatorial structure of a household grows far faster than its headcount. Neither tradition asks the question taken up here. The question is not how many relationships or subgroups a family contains. It is how many distinct ways the children can divide into opposing sides, and how that number scales.

2. The combinatorial framework

Game theorists and multi-agent systems researchers have studied coalition structure generation extensively. Their central question is how many ways a set of n agents can be partitioned into cooperating or opposing groups, and how to search that space efficiently [12, 11]. The framework transfers directly to sibling groups once each child is treated as an agent who can be on one side of a conflict, on the opposing side, or uninvolved.

A conflict configuration is then an unordered pair of disjoint, nonempty subsets of the n children. The two subsets are the sides; any remaining children sit it out. Each child can be assigned to side A , side B , or neither, giving 3^n assignments. Inclusion-exclusion removes the $2 \cdot 2^n - 1$ assignments in which a side is empty, and dividing by two accounts for the interchangeable labels A and B :

$$C(n) = \frac{3^n - 2 \cdot 2^n + 1}{2}. \quad (1)$$

Concretely, take three children named A, B, and C. The six configurations are the three one-on-one disputes (A against B with C uninvolved, and its two counterparts) and the three two-against-one splits ($\{A, B\}$ against C, and its two counterparts).

The expression in equation (1) is not ad hoc. It equals $S(n+1, 3)$, a Stirling number of the second kind, and it appears in the On-Line Encyclopedia of Integer Sequences as A000392, where one of its listed characterizations is exactly this: the number of ways to form disjoint unions of two nonempty subsets of an n -element set [10]. Table 1 collects the values for $n = 1$ through 8 next to Bossard’s dyadic count. Figure 1 plots the range where most families actually live, $n \leq 4$, with every value readable. Figure 2 extends the range to $n = 8$, where the shape of the growth takes over.

Table 1: Possible conflict configurations, $C(n)$ of equation (1), against pairwise relationships, $n(n-1)/2$. All values are exact.

Children n	Conflict configurations $C(n)$	Sibling pairs $n(n-1)/2$
1	0	0
2	1	1
3	6	3
4	25	6
5	90	10
6	301	15
7	966	21
8	3,025	28

Four children allow 25 possible conflict configurations, but form only 6 sibling pairs

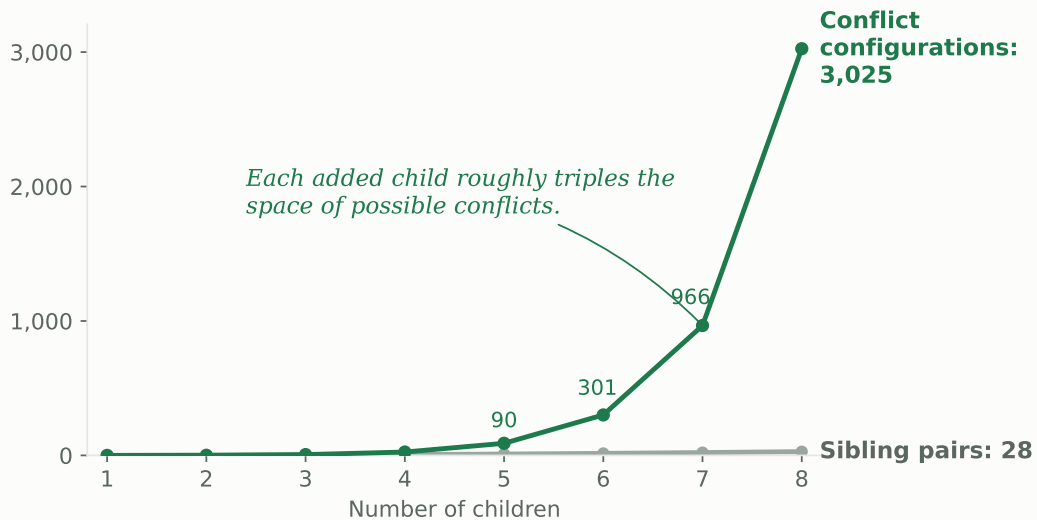
Ways n children can split into two opposing sides, others uninvolved: $(3^n - 2 \cdot 2^n + 1)/2$, vs. sibling pairs, $n(n-1)/2$ · a possibility space, no claim about actual conflicts



Figure 1: Conflict configurations $C(n) = (3^n - 2 \cdot 2^n + 1)/2$, the ways n children can split into two opposing sides with the remaining children uninvolved, against pairwise relationships $n(n-1)/2$, for the family sizes most readers actually have ($n = 1$ to 4). The two counts agree at one and two children and diverge from three onward: 6 configurations against 3 pairs at $n = 3$, then 25 against 6 at $n = 4$. The counts are a possibility space, exact by construction. They carry no claim about the frequency or severity of actual conflicts.

Eight children allow 3,025 possible conflict configurations, but form only 28 sibling pairs

Ways n children can split into two opposing sides, others uninvolved: $(3^n - 2 \cdot 2^n + 1)/2$, vs. sibling pairs, $n(n-1)/2$ · a possibility space, no claim about actual conflicts



Source: OEIS A000392, $S(n+1, 3)$ · computed 2026-08-12 · exact counts, nothing estimated

Figure 2: The same two counts extended to $n = 8$. Successive values of $C(n)$ approach a constant ratio of 3 (90, 301, 966, 3,025), so the growth is exponential, while the dyadic count below it is quadratic and reaches only 28. The values of Figure 1 sit in the flat left edge of this frame. As before, the counts are a possibility space, not observed conflicts.

The shape of the growth is the point. The jump from two children to three is six-fold. The jump from three to four multiplies the count by roughly four again. Nothing linear behaves this way, and no quadratic does either; the growth rate tracks 3^n . Bossard’s dyadic count, by contrast, is quadratic. His framework registers the difference between two children and eight as 1 relationship versus 28. The conflict-configuration space registers it as 1 versus 3,025.

One note on related formalisms. A similar ally/opponent/neutral trichotomy underlies “three-way conflict analysis” in the tradition of Pawlak’s conflict models, which has been applied to labor and political disputes [7]. That literature analyzes the structure of a given conflict. This article counts the space of structures available.

3. Why this matters, and what this article does not claim

Whether the number of possible conflict configurations determines actual conflict frequency, severity, or long-term relational outcomes is an empirical question, and nothing here settles it. Many families with high theoretical conflict potential remain close. Some two-child families fracture. What the framework offers is a way to quantify the possibility space that parents, therapists, and researchers are implicitly reacting to when they describe larger families as harder to manage or more prone to shifting alliances.

The empirical literature gives the reframing some initial traction. Sibling estrangement in adulthood is common enough to measure at scale; in German panel data, 28% of respondents reported at least one episode of estrangement from a sibling [4]. Recent work on medium-to-large families finds that sibling relationship quality in larger sibships is multifaceted, with substantial variation across dyads within the same family [5]. Within-family variation across dyads is what a large configuration space would predict, though it is consistent with other explanations too. The concrete proposal for family cohesion research is this: test whether reported conflict frequency, coalition instability, or estrangement outcomes correlate with the theoretical conflict space of a family, independent of family size alone.

4. Directions for further research

Researchers with access to sibling relationship data could test whether the theoretical conflict space correlates with reported measures of sibling closeness, conflict frequency, or estrangement in adulthood. Another open question is whether particular family structures matter: evenly split coalitions are available to four-child families, for example, while three-child families are limited to one-against-one and two-against-one forms, and those structural differences might correlate with different relational outcomes. The framework could also be extended to model coalition stability over time, since sibling alliances are known to shift, and the count developed here is static; it says which configurations are possible, not which ones occur or persist. The

coalition structure generation literature has built tools for reasoning about exactly such spaces [11] and may supply the machinery for that extension.

Conclusion

The number of possible sibling conflict configurations grows exponentially with family size, not linearly. Why some large families remain close while others fracture is a question this article leaves alone. What it offers is a map of the terrain: a measure of how much structural complexity a family navigates as it grows. Bossard and Kephart began the mapping three-quarters of a century ago with relationships and subgroups. Extending it to opposing-sides configurations reveals a possibility space vastly larger still. I offer the framework in the hope that it proves useful to parents seeking context for what they are experiencing, and to researchers who want to test whether this structural complexity is itself a meaningful variable in family cohesion outcomes.

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